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# Conservative oscillations and perturbation

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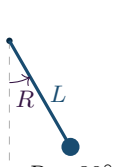
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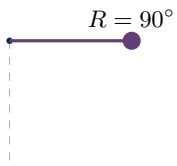
2.7 Validation, limitations and solved practice

# Does a larger pendulum swing take longer?

Two releases from rest have the same length $L$ (m) and gravitational acceleration $g$ (m/s$^2$), but different initial angles $R$. The ratios below come from the exact ideal-pendulum period integral derived next.




$T/T_0 = 1.0174$




$T/T_0 = 1.1803$

The larger release takes longer. The linear period $T_0 = 2\pi\sqrt{L/g}$ has no amplitude in it, so it cannot explain this difference.

Conserved energy relates the release angle $R$ to the time taken for one swing.

# Energy gives the period without solving the motion

$$m\ddot{x} + \frac{\mathrm{d}U}{\mathrm{d}x} = 0, \qquad E = \tfrac{1}{2}m\dot{x}^2 + U(x).$$

For mass $m$ (kg) and potential energy $U$ (J), multiply by $\dot{x}$: $m\ddot{x}\dot{x} + (\mathrm{d}U/\mathrm{d}x)\dot{x} = \mathrm{d}E/\mathrm{d}t = 0$.

The turning points $x_-$ and $x_+$ satisfy $U(x_-) = U(x_+) = E$. Between them $E - U(x) > 0$, so the velocity is real.

**On the rightward branch**

$$\dot{x} = \sqrt{\frac{2}{m}[E - U(x)]}, \qquad \mathrm{d}t = \sqrt{\frac{m}{2}}\,\frac{\mathrm{d}x}{\sqrt{E - U(x)}}.$$

Travel from $x_-$ to $x_+$ takes half a cycle. Double its time integral.

$$\boxed{T(E) = \sqrt{2m}\int_{x_-}^{x_+} \frac{\mathrm{d}x}{\sqrt{E - U(x)}}, \qquad \omega(E) = \frac{2\pi}{T(E)}.}$$

# Worked example: recover the linear period

A linear spring has stiffness $k$ (N/m); release the mass from rest at $x = R$ (m).

$$U(x) = \tfrac{1}{2}kx^2, \qquad E = \tfrac{1}{2}kR^2, \qquad x_\pm = \pm R.$$

$$T = 2\sqrt{m/k}\int_{-R}^{R} \frac{\mathrm{d}x}{\sqrt{R^2 - x^2}}.$$

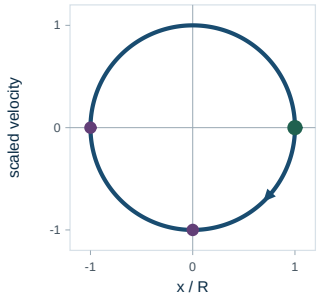
With $x = R\sin\psi$, the limits become $-\pi/2, \pi/2$ and $\mathrm{d}x/\sqrt{R^2 - x^2} = R\cos\psi\,\mathrm{d}\psi/(R\cos\psi) = \mathrm{d}\psi$.

$$T = 2\sqrt{m/k}\int_{-\pi/2}^{\pi/2} \mathrm{d}\psi = \boxed{2\pi\sqrt{m/k}.}$$

The amplitude cancels. Thus $\omega_0 = \sqrt{k/m}$ is independent of $R$ for this linear model.

# A closed phase orbit takes one period

For the linear spring, $x/R = \cos(\omega_0 t)$ and $v/(\omega_0 R) = -\sin(\omega_0 t)$, where $v = \dot{x}$. This recalls the Week 1 phase portrait and adds travel time.




**Release: x = R, v = 0**

First quarter: x = 0

Half-cycle: x = -R

Return to R: one period T

**Time runs clockwise.**

Starting at $(x, v) = (R, 0)$, the orbit reaches $x = 0$ after $T/4$ and $x = -R$ after $T/2$. One closed lap takes $T$, the time measured by the energy integral.

# Pendulum: release energy gives the angular speed

$$\theta(0) = R, \quad \dot{\theta}(0) = 0, \quad 0 < R < \pi, \qquad \omega_0 = \sqrt{g/L}.$$

The model is $\ddot{\theta} + \omega_0^2 \sin\theta = 0$. Multiplication by $\dot{\theta}$ gives a total time derivative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\tfrac{1}{2}\dot{\theta}^2 + \omega_0^2(1 - \cos\theta)\right] = 0.$$

At release $\dot{\theta} = 0$, so the conserved quantity equals $\omega_0^2(1 - \cos R)$:

$$\tfrac{1}{2}\dot{\theta}^2 + \omega_0^2(1 - \cos\theta) = \omega_0^2(1 - \cos R).$$

$$\boxed{\dot{\theta}^2 = 2\omega_0^2(\cos\theta - \cos R).}$$

On the first quarter-cycle $\theta$ decreases from $R$ to 0. Use the negative square root for $\dot{\theta}$ when converting $\mathrm{d}t = \mathrm{d}\theta/\dot{\theta}$.

# Pendulum: substitute into the quarter-period integral

During the first quarter-cycle $\theta$ falls from $R$ to 0, so $\dot{\theta} = -\omega_0\sqrt{2(\cos\theta - \cos R)}$. With $\mathrm{d}t = \mathrm{d}\theta/\dot{\theta}$, reverse the limits to make the time positive:

$$\frac{T}{4} = \int_R^0 \frac{\mathrm{d}\theta}{-\omega_0\sqrt{2(\cos\theta - \cos R)}}.$$

$$\frac{T}{4} = \frac{1}{\omega_0}\int_0^R \frac{\mathrm{d}\theta}{\sqrt{2(\cos\theta - \cos R)}}.$$

This is the time for one quarter of the orbit. Symmetry gives four equal quarters.

# Pendulum: transform the period integral

Start from $T = 4\omega_0^{-1} \int_0^R [2(\cos\theta - \cos R)]^{-1/2}\, \mathrm{d}\theta$.

Set $\sin(\theta/2) = \kappa \sin\psi$, where $\kappa = \sin(R/2)$. The new limits are 0 and $\pi/2$.

$$2(\cos\theta - \cos R) = 4\kappa^2 \cos^2\psi, \qquad \mathrm{d}\theta = \frac{2\kappa\cos\psi}{\sqrt{1-\kappa^2\sin^2\psi}}\,\mathrm{d}\psi.$$

Cancel $2\kappa\cos\psi$ between numerator and denominator. Four equal quarter-cycles give $T$.

$$\boxed{T = \frac{4}{\omega_0} K(\kappa), \quad K(\kappa) = \int_0^{\pi/2} \frac{\mathrm{d}\psi}{\sqrt{1-\kappa^2\sin^2\psi}}.}$$

$\kappa$ is the elliptic modulus; some software instead accepts the parameter $\kappa^2$. The integral is exact for the ideal pendulum.

# Pendulum: expand the period at small amplitude

For small release angle $R$, the modulus $\kappa = \sin(R/2)$ is small. Expand the integrand of $K(\kappa)$ by setting $z = \kappa^2 \sin^2\psi$ in the binomial series:

$$(1-z)^{-1/2} = 1 + \tfrac{1}{2}z + \tfrac{3}{8}z^2 + \cdots, \qquad K = \frac{\pi}{2}\left(1 + \frac{\kappa^2}{4} + \frac{9\kappa^4}{64} + \cdots\right).$$

Integrate powers term by term: $\int_0^{\pi/2} \sin^2\psi\,\mathrm{d}\psi = \pi/4$ and $\int_0^{\pi/2} \sin^4\psi\,\mathrm{d}\psi = 3\pi/16$.

Use $\kappa^2 = R^2/4 - R^4/48 + O(R^6)$ and $\kappa^4 = R^4/16 + O(R^6)$; collect equal powers of $R$.

$$\boxed{\frac{T}{T_0} = 1 + \frac{R^2}{16} + \frac{11R^4}{3072} + O(R^6)}, \qquad T_0 = \frac{2\pi}{\omega_0}.$$

$$\frac{\omega}{\omega_0} = 1 - \frac{R^2}{16} + \frac{R^4}{3072} + O(R^6).$$

As $R$ increases within the range where this small-angle series is accurate, the period increases and the frequency decreases. The exact integral shows the same trend at larger angles.

# Pendulum: calculate and compare

For $L = 1$ m and $g = 9.81$ m/s$^2$, $T_0 = 2.006067$ s.

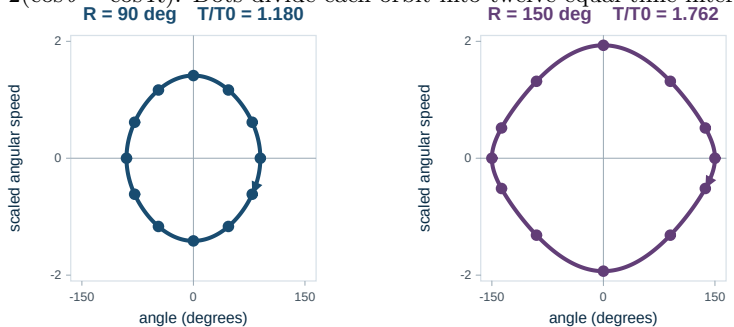
| $R$ | Exact $T/T_0$ | Exact $T$ (s) | $1 + R^2/16$ |
|---|---|---|---|
| 30° | 1.017409 | 2.040990 | 1.017135 |
| 60° | 1.073182 | 2.152875 | 1.068539 |
| 90° | 1.180341 | 2.367842 | 1.154213 |

At 60°, convert to $R = \pi/3$ rad before using $1 + R^2/16 = 1.068539$.

$T \approx 2.006067(1.068539) = 2.14356$ s, about 0.433% below the exact ideal-pendulum period 2.152875 s. The leading correction is useful here, but its error grows with $R$.

# A larger pendulum orbit takes a longer lap

For each release, $(\dot{\theta}/\omega_0)^2 = 2(\cos\theta - \cos R)$. Dots divide each orbit into twelve equal time intervals.




The 150° swing moves faster at $\theta = 0$ yet takes $1.762T_0$ per lap, versus $1.180T_0$ at 90°. Near $R = \pi$, slowing by the turning angle makes the period grow sharply.

# One free orbit gives one backbone point

**Release from rest at $x(0) = R$: follow that orbit to its frequency**

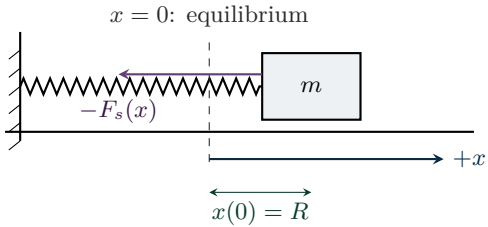
$$R = x_+ \;\longrightarrow\; E = U(R) \;\longrightarrow\; T(E) \;\longrightarrow\; (R, \omega), \qquad \omega = \frac{2\pi}{T(E)}.$$

Repeat for several releases $R$ and plot the resulting points $(R, \omega)$. The curve through them is the free-oscillation backbone. Here its amplitude axis means release displacement $R$.

$$\frac{\mathrm{d}\omega}{\mathrm{d}R} > 0 : \text{ hardening}, \qquad \frac{\mathrm{d}\omega}{\mathrm{d}R} < 0 : \text{ softening}.$$

Here $R = x(0) > 0$ is the release displacement and $\dot{x}(0) = 0$. The backbone plots frequency against this $R$.

# Spring model: displacement, release and restoring force



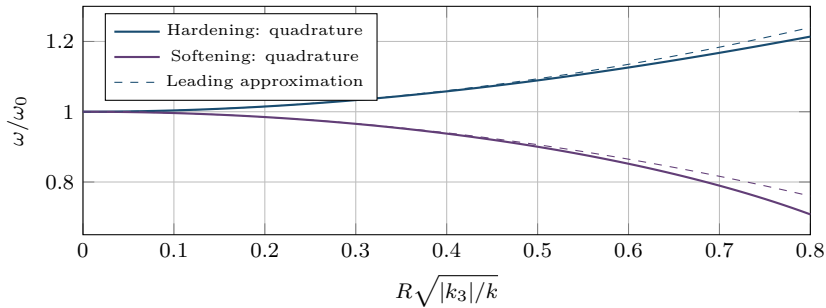

$$F_s(x) = kx + k_2x^2 + k_3x^3$$
$$m\ddot{x} + F_s(x) = 0$$
Release from rest: $\dot{x}(0) = 0$

$m$: mass (kg); $k$: linear stiffness (N/m); $k_2$: quadratic stiffness (N/m$^2$); $k_3$: cubic stiffness (N/m$^3$). Displacement $x$ and release $R$ are in metres.

Pure cubic spring: $k_2 = 0$. Pure quadratic spring: $k_3 = 0$. The actual spring force on the mass is $-F_s(x)$; the arrow shows its direction for positive restoring force.

# Hardening and softening backbones

For $m\ddot{x} + kx + k_3x^3 = 0$, the horizontal axis is the release $R$ scaled by $\sqrt{|k_3|/k}$. Positive $k_3$ hardens; negative $k_3$ softens.




Solid: period from the energy integral. Dashed: the small-amplitude approximation derived in 2.4. At the same release $R$, positive $k_3$ raises the frequency and negative $k_3$ lowers it.

# Cubic spring: scale displacement and time

For $m\ddot{x} + kx + k_3x^3 = 0$, release from rest at $x(0) = R$. Define

$$u = \frac{x}{R}, \qquad \omega_0 = \sqrt{\frac{k}{m}}, \qquad \tau = \omega_0 t.$$

Primes denote derivatives with respect to $\tau$. The chain rule gives

$$\dot{x} = R\omega_0 u', \qquad \ddot{x} = R\omega_0^2 u''.$$

Substitute these into the equation, use $m\omega_0^2 = k$, and divide by $kR$:

$$kRu'' + kRu + k_3R^3u^3 = 0 \quad \Longrightarrow \quad u'' + u + \frac{k_3R^2}{k}u^3 = 0.$$

$$\boxed{\varepsilon = \frac{k_3R^2}{k}, \qquad u'' + u + \varepsilon u^3 = 0, \qquad u(0) = 1, \quad u'(0) = 0.}$$

$\varepsilon$ is the cubic-to-linear restoring-force ratio at release. Its sign distinguishes hardening from softening; perturbation requires $|\varepsilon| \ll 1$.

# Cubic spring: period from conserved energy

Conserved energy gives a period for the cubic-spring model at any bounded release.

$$m\ddot{x} + kx + k_3x^3 = 0, \quad x = Ru, \quad \tau = \sqrt{k/m}\,t, \quad \varepsilon = k_3R^2/k.$$

$$u'' + u + \varepsilon u^3 = 0, \quad u(0) = 1, \quad u'(0) = 0.$$

Multiply the equation by $u'$ and integrate. At release $u = 1$, $u' = 0$, so

$$\tfrac{1}{2}u'^2 + \tfrac{1}{2}u^2 + \tfrac{\varepsilon}{4}u^4 = \tfrac{1}{2} + \tfrac{\varepsilon}{4}.$$

Rearrange and factor the velocity squared:

$$u'^2 = (1 - u^2)\left[1 + \frac{\varepsilon}{2}(1 + u^2)\right].$$

On the first quarter-cycle $u$ decreases from 1 to 0. The energy equation gives its speed without solving for $u(\tau)$.

# Turn the cubic-spring speed into a period

On the first quarter-cycle take the negative root of $u'^2$ and use $\mathrm{d}\tau = \mathrm{d}u/u'$. Reversing the limits gives

$$\frac{T_\tau}{4} = \int_0^1 \frac{\mathrm{d}u}{\sqrt{(1-u^2)[1+\frac{\varepsilon}{2}(1+u^2)]}}.$$

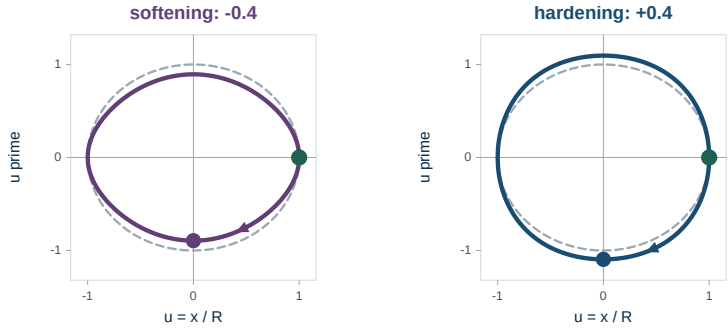
Set $u = \sin\psi$, so $\mathrm{d}u = \cos\psi\,\mathrm{d}\psi$ and $\sqrt{1-u^2} = \cos\psi$ for $0 \le \psi \le \pi/2$. They cancel.

$$\boxed{T_\tau = 4\int_0^{\pi/2} \frac{\mathrm{d}\psi}{\sqrt{1+\frac{\varepsilon}{2}(1+\sin^2\psi)}}}, \quad \Omega = \frac{\omega}{\omega_0} = \frac{2\pi}{T_\tau}.$$

This integral is exact for the stated cubic-spring model and supplies an independent check of the perturbation approximation.

# At the same release, cubic stiffness changes the orbit

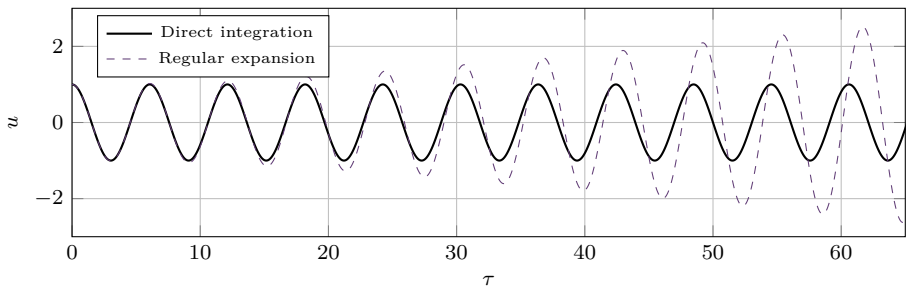
For $u'' + u + \varepsilon u^3 = 0$, compare $\varepsilon = -0.4$ and $+0.4$ at the same release $u = 1$ from rest. The dashed loop is the linear case.




Energy gives $|u'| = \sqrt{1 + \varepsilon/2}$ at $u = 0$: 0.894 for $\varepsilon = -0.4$ and 1.095 for $+0.4$. The full-orbit integral gives $T/T_0 = 1.200$ and 0.878, respectively.

# Can a bounded oscillator have a growing approximation?

$$u'' + u + \varepsilon u^3 = 0, \qquad \varepsilon = 0.10, \qquad u(0) = 1, \quad u'(0) = 0.$$




$\varepsilon = 0.10$ and release $(u, u') = (1, 0)$. The first-order approximation drifts because it keeps the linear frequency; the coming method corrects that frequency.

# Expand the displacement at fixed linear time

$$u'' + u + \varepsilon u^3 = 0, \quad u(0) = 1, \quad u'(0) = 0, \quad |\varepsilon| \ll 1.$$

$$u = u_1 + \varepsilon u_2 + O(\varepsilon^2).$$

Subscripts number successive terms. Since $u = x/R$ already has unit release, its leading term is $u_1$. Substitute $\varepsilon u^3 = \varepsilon u_1^3 + O(\varepsilon^2)$ and match powers of $\varepsilon$:

$$O(1): \quad u_1'' + u_1 = 0, \quad u_1(0) = 1, \quad u_1'(0) = 0,$$

$$O(\varepsilon): \quad u_2'' + u_2 = -u_1^3, \quad u_2(0) = u_2'(0) = 0.$$

$u_1 = \cos\tau$. Primes mean $\mathrm{d}/\mathrm{d}\tau$, where $\tau = \omega_0 t$ and $\omega_0 = \sqrt{k/m}$.

# Solve the equation for the next term

$$u_2'' + u_2 = -\cos^3\tau = -\tfrac{3}{4}\cos\tau - \tfrac{1}{4}\cos 3\tau.$$

The right-hand side has two cosines. To solve for each part, direct differentiation gives

$$(\cos 3\tau)'' + \cos 3\tau = -8\cos 3\tau, \quad (\tau\sin\tau)'' + \tau\sin\tau = 2\cos\tau.$$

Therefore a particular solution is $-3\tau\sin\tau/8 + \cos 3\tau/32$. Add $C\cos\tau + S\sin\tau$. The conditions $u_2(0) = u_2'(0) = 0$ give $C = -1/32$ and $S = 0$.

$$\boxed{u_2 = -\frac{3}{8}\tau\sin\tau + \frac{\cos 3\tau - \cos\tau}{32}.}$$

The resonant term $-3\tau\sin\tau/8$ is secular. Its contribution becomes order one when $|\varepsilon|\tau = O(1)$. This is approximation failure, not physical instability.

# A small frequency shift creates the secular term

The term $-3\varepsilon\tau\sin\tau/8$ in $\varepsilon u_2$ comes from expanding a cosine with a slightly changed frequency:

$$\cos\left[\left(1+\frac{3\varepsilon}{8}\right)\tau\right] = \cos\tau\cos\frac{3\varepsilon\tau}{8} - \sin\tau\sin\frac{3\varepsilon\tau}{8}.$$

When $|\varepsilon\tau| \ll 1$, use $\cos(3\varepsilon\tau/8) \simeq 1$ and $\sin(3\varepsilon\tau/8) \simeq 3\varepsilon\tau/8$. This reproduces $\cos\tau - 3\varepsilon\tau\sin\tau/8$.

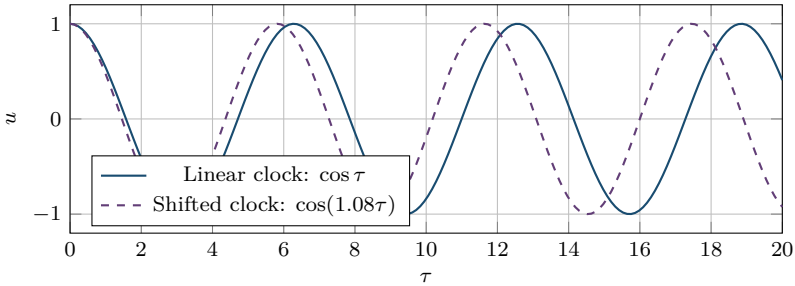
For $\varepsilon = 0.10$, ten linear periods mean $\tau = 20\pi$. The first-order phase change is

$$\tau = 20\pi, \quad \Delta\phi \approx \frac{3\varepsilon}{8}\tau = \boxed{2.35619 \text{ rad} = 135^\circ.}$$

This large $\Delta\phi$ is beyond the small-phase expansion. It shows why the truncated regular series loses accuracy after many cycles; the motion itself remains bounded.

# The oscillator needs a corrected clock

The motion stays bounded, but a small frequency error grows into a visible phase error. The 8% shift below illustrates the idea; it is not a fitted model result.




After time $\tau$, the phase difference is $0.08\tau$. Poincaré–Lindstedt expands the frequency along with the waveform, so the approximation uses the corrected clock.

# Use a phase that follows the unknown frequency

In $u'' + u + \varepsilon u^3 = 0$, primes mean derivatives in $\tau = \omega_0 t$. Describe the motion using the new phase $\phi = \Omega\tau$, where $\Omega = \omega/\omega_0$ is unknown. Expand it as

$$\Omega = 1 + \varepsilon\Omega_1 + O(\varepsilon^2).$$

The chain rule gives $\mathrm{d}\phi/\mathrm{d}\tau = \Omega$. Since $\Omega$ is constant for a given motion,

$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = \Omega u_\phi, \qquad \frac{\mathrm{d}^2u}{\mathrm{d}\tau^2} = \Omega^2 u_{\phi\phi}.$$

$$\boxed{\Omega^2 u_{\phi\phi} + u + \varepsilon u^3 = 0}, \qquad \Omega^2 = 1 + 2\varepsilon\Omega_1 + O(\varepsilon^2).$$

One cycle is $2\pi$ in $\phi$, so its dimensionless period is $T_\tau = 2\pi/\Omega$. The unknown $\Omega_1$ can remove the secular term.

# Collect the first-order terms

Insert $u = u_1 + \varepsilon u_2 + \cdots$ and $\Omega^2 = 1 + 2\varepsilon\Omega_1 + \cdots$ into $\Omega^2 u_{\phi\phi} + u + \varepsilon u^3 = 0$.

$$O(1): \quad u_{1,\phi\phi} + u_1 = 0, \quad u_1(0) = 1, \quad u_{1,\phi}(0) = 0 \quad \Longrightarrow \quad u_1 = \cos\phi.$$

At $O(\varepsilon)$, the acceleration contributes $u_{2,\phi\phi} + 2\Omega_1 u_{1,\phi\phi}$. The spring terms contribute $u_2 + u_1^3$. Thus

$$u_{2,\phi\phi} + u_2 = -u_1^3 - 2\Omega_1 u_{1,\phi\phi}.$$

Use $u_{1,\phi\phi} = -\cos\phi$ and $\cos^3\phi = (3\cos\phi + \cos 3\phi)/4$:

$$u_{2,\phi\phi} + u_2 = \left(2\Omega_1 - \frac{3}{4}\right)\cos\phi - \frac{1}{4}\cos 3\phi.$$

# Remove the resonant term and satisfy release data

A $\cos\phi$ term on the right would produce $\phi\sin\phi$. Set its coefficient to zero: $2\Omega_1 - 3/4 = 0$. Then

$$\boxed{\Omega_1 = \frac{3}{8}}, \qquad u_{2,\phi\phi} + u_2 = -\tfrac{1}{4}\cos 3\phi.$$

The $\cos 3\phi$ part gives $\cos 3\phi/32$. Add a free $\cos\phi$ term so $u_2(0) = 0$; the zero initial velocity removes the sine term.

$$\boxed{u_2 = \frac{\cos 3\phi - \cos\phi}{32}}.$$

$$u \approx \cos\phi + \frac{\varepsilon}{32}(\cos 3\phi - \cos\phi), \qquad \phi = \left(1 + \frac{3\varepsilon}{8}\right)\tau.$$

This approximation stays periodic. The correction is zero at $\phi = 0$, so it preserves release from $u(0) = 1$.

# Worked example: a hardening spring

$$m = 1 \text{ kg}, \quad k = 100 \text{ N/m}, \quad k_3 = 1000 \text{ N/m}^3, \quad R = 0.10 \text{ m}.$$

First find the linear frequency and the dimensionless nonlinear strength at the stated release:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{1}} = 10 \text{ rad/s}, \qquad \varepsilon = \frac{k_3 R^2}{k} = \frac{1000(0.10)^2}{100} = 0.10.$$

Then apply $\Omega = 1 + 3\varepsilon/8$ and convert back to physical time:

$$\Omega_{\mathrm{PL}} = 1 + \frac{3(0.10)}{8} = 1.0375, \qquad \omega_{\mathrm{PL}} = \omega_0 \Omega_{\mathrm{PL}} = 10(1.0375) = 10.375 \text{ rad/s}.$$

$$\boxed{T_{\mathrm{PL}} = \frac{2\pi}{\omega_{\mathrm{PL}}} = \frac{2\pi}{10.375} = 0.605608 \text{ s}.}$$

Exact-model quadrature: $\omega = 10.36716907$ rad/s and $T = 0.606065674$ s. The first-order frequency error is 0.075536%.

# Hardening example: check the harmonic amplitudes

From the preceding result, $\phi = \omega_{\mathrm{PL}} t = 10.375t$ and $\varepsilon/32 = 0.10/32 = 0.003125$.

$$x(t) \approx 0.10\,[\cos(10.375t) + 0.003125\{\cos(31.125t) - \cos(10.375t)\}] \ \text{m},$$
$$= 0.0996875\cos(10.375t) + 0.0003125\cos(31.125t) \ \text{m}.$$

The fundamental component is slightly smaller than the release displacement. The third harmonic supplies the difference when the two cosines equal one at $t = 0$.

$$\boxed{x(0) = 0.0996875 + 0.0003125 = R = 0.10 \ \text{m}, \qquad \dot{x}(0) = 0.}$$

A good frequency prediction does not guarantee accurate phase for arbitrarily long times. An omitted $O(\varepsilon^2)$ frequency term can produce $O(\varepsilon^2 \tau)$ phase drift.

# Pendulum: recover the softening correction

Start with $\ddot{\theta} + (g/L)\sin\theta = 0$. With $\theta = Ru$ and $\tau = \sqrt{g/L}\,t$, divide by $R(g/L)$:

$$\theta = Ru, \quad \tau = \sqrt{g/L}\,t, \qquad u'' + \frac{\sin(Ru)}{R} = 0.$$

Expand $\sin(Ru) = Ru - R^3u^3/6 + O(R^5)$ and divide by $R$.

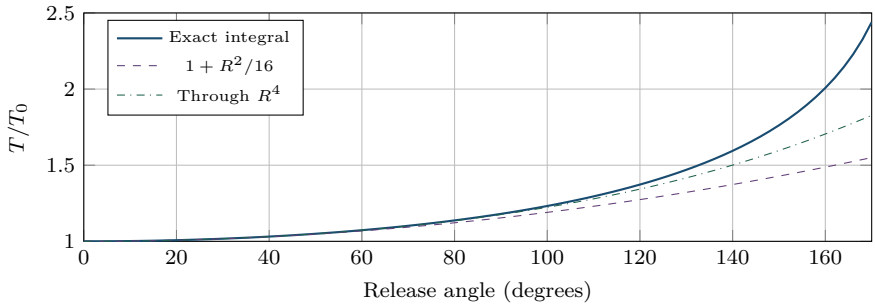
$$u'' + u - \frac{R^2}{6}u^3 + O(R^4) = 0.$$

Here $\varepsilon = -R^2/6$. Insert it into $\Omega = 1 + 3\varepsilon/8 + O(\varepsilon^2)$.

$$\varepsilon = -R^2/6, \qquad \boxed{\omega/\omega_0 = 1 - R^2/16 + O(R^4).}$$

At $R = 30^\circ$, PL gives 0.9828653; the exact-model ratio is 0.9828891. This agrees with the energy-integral expansion. Use radians.

# Pendulum: know when the small-angle expansion fails




Near $R = \pi$, the exact period diverges. To obtain the correct $R^4$ frequency term, retain both the next perturbation order and the quintic sine term.

# Why can the left and right turns differ?

Here $k_2$ is quadratic stiffness ($\mathrm{N/m^2}$). In $m\ddot{x} + kx + k_2x^2 = 0$, the term $k_2x^2$ has the same sign on either side of $x = 0$. For $k_2 > 0$ and small $|x|$, the restoring force is stronger at $x > 0$ and weaker at $x < 0$.

$$x = Ru, \quad \tau = \sqrt{k/m}\, t, \quad \delta = \frac{k_2 R}{k}.$$

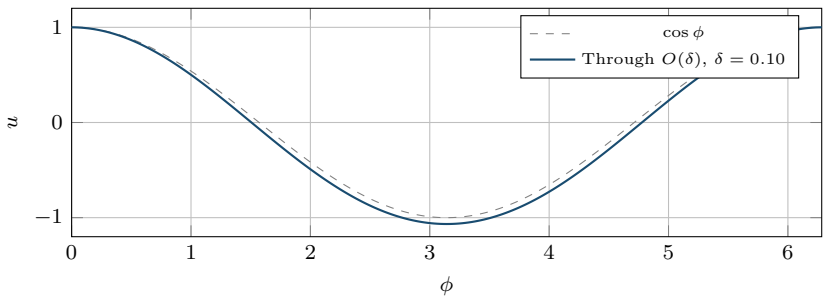
Substitute $\ddot{x} = R(k/m)u''$ and divide the equation by $kR$:

$$\boxed{u'' + u + \delta u^2 = 0}, \qquad u(0) = 1, \quad u'(0) = 0.$$

Here $R$ is the positive release from rest, primes mean derivatives in $\tau$, and $|\delta| \ll 1$ for the perturbation calculation that follows.

# Quadratic stiffness makes a cycle asymmetric

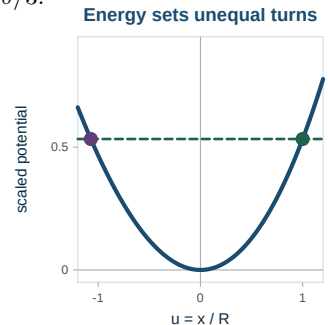
$$u'' + u + \delta u^2 = 0, \qquad \delta = 0.10, \qquad u(0) = 1, \quad u'(0) = 0.$$

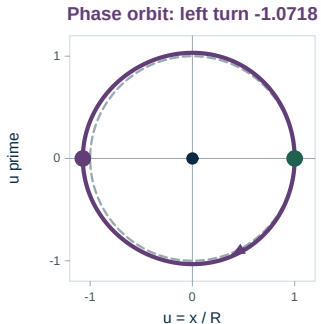



This is displacement $u$ against phase $\phi = \Omega\tau$ over one cycle, not a phase-plane plot. The turning points are $+1$ and about $-1.072$; the mean is about $-0.05$.

# Potential and phase orbit show the same asymmetry

For $u'' + u + \delta u^2 = 0$ with $\delta = 0.10$, release at $u = 1$ gives the scaled potential $V(u) = u^2/2 + \delta u^3/3$ and energy level $V(1) = 1/2 + \delta/3$.




The energy line and phase orbit share turning points $u = 1$ and $u = -1.0718$. Equilibrium remains at $u = 0$. The negative mean found next is the time average of this asymmetric cycle.

# Expand the phase and waveform

For the scaled model $u'' + u + \delta u^2 = 0$, use the phase $\phi = \Omega\tau$ and expand both waveform and frequency:

$$u = u_1 + \delta u_2 + \delta^2 u_3 + \cdots, \qquad \Omega = 1 + \delta\Omega_1 + \delta^2\Omega_2 + \cdots.$$

As in the cubic case, $\mathrm{d}^2u/\mathrm{d}\tau^2 = \Omega^2 u_{\phi\phi}$. Insert the expansions and collect the terms independent of $\delta$:

$$\Omega^2 u_{\phi\phi} + u + \delta u^2 = 0, \qquad O(1):\ u_{1,\phi\phi} + u_1 = 0 \implies u_1 = \cos\phi.$$

The leading term already satisfies release, so $u_2(0) = u_{2,\phi}(0) = 0$.

# The first correction has no resonant term

Collect the $\delta$ terms in $\Omega^2 u_{\phi\phi} + u + \delta u^2 = 0$:

$$u_{2,\phi\phi} + u_2 = -u_1^2 - 2\Omega_1 u_{1,\phi\phi}.$$

Since $u_1 = \cos\phi$, use $u_{1,\phi\phi} = -\cos\phi$ and $\cos^2\phi = (1 + \cos 2\phi)/2$:

$$u_{2,\phi\phi} + u_2 = 2\Omega_1 \cos\phi - \tfrac{1}{2}(1 + \cos 2\phi).$$

The quadratic term gives a constant and $\cos 2\phi$, but no $\cos\phi$. Set $2\Omega_1 = 0$ so the right side has no resonant $\cos\phi$ term.

$$\boxed{\Omega_1 = 0}, \qquad u_{2,\phi\phi} + u_2 = -\tfrac{1}{2} - \tfrac{1}{2}\cos 2\phi.$$

Quadratic stiffness changes the waveform at first order. Its frequency shift begins at the next order.

# Solve the first correction: mean shift and second harmonic

$$u_{2,\phi\phi} + u_2 = -\tfrac{1}{2}(1 + \cos 2\phi).$$

A constant $-1/2$ solves the constant part. Since $(\cos 2\phi)_{\phi\phi} + \cos 2\phi = -3\cos 2\phi$, the second-harmonic part is $\cos 2\phi/6$.

Add a free $C\cos\phi + S\sin\phi$. At release, $0 = u_2(0) = -1/2 + 1/6 + C$ and $0 = u_{2,\phi}(0) = S$; thus $C = 1/3$, $S = 0$.

$$\boxed{u_2 = -\frac{1}{2} + \frac{1}{3}\cos\phi + \frac{1}{6}\cos 2\phi.}$$

The mean is $\bar{u} = -\delta/2 + O(\delta^2)$. At $\phi = \pi$, $u_2 = -2/3$, so the negative turning point is $u_- = -1 - 2\delta/3 + O(\delta^2)$.

# The frequency shift appears at second order

Now $\Omega_1 = 0$ and $\Omega^2 = 1 + 2\delta^2\Omega_2 + \cdots$. The $\delta^2$ terms give

$$u_{3,\phi\phi} + u_3 = -2u_1u_2 - 2\Omega_2 u_{1,\phi\phi}.$$

Insert $u_1 = \cos\phi$ and $u_2 = -1/2 + (1/3)\cos\phi + (1/6)\cos 2\phi$. Product-to-sum gives

$$-2u_1u_2 = \tfrac{5}{6}\cos\phi - \tfrac{1}{3} - \tfrac{1}{3}\cos 2\phi - \tfrac{1}{6}\cos 3\phi.$$

Because $u_{1,\phi\phi} = -\cos\phi$, the frequency term adds $2\Omega_2\cos\phi$. Remove the resonant cosine by setting

$$2\Omega_2 + \frac{5}{6} = 0 \quad \Longrightarrow \quad \boxed{\Omega = 1 - \frac{5}{12}\delta^2 + O(\delta^3).}$$

The leading frequency contribution is softening for either sign of $k_2$. It comes from feedback of the first-order waveform.

# Worked example: what does $\delta = 0.10$ predict?

$$u'' + u + 0.10u^2 = 0, \qquad u(0) = 1, \quad u'(0) = 0.$$

From the preceding calculations, use the first-order waveform for displacement and the second-order correction for frequency:

$$\Omega_{\mathrm{PL}} = 1 - \frac{5}{12}(0.10)^2 = 0.99583333, \qquad \bar{u} \approx -\frac{0.10}{2} = -0.0500.$$

At the negative turning point $\phi = \pi$, $u_1 = -1$ and $u_2 = -2/3$, so

$$\boxed{u_- \approx -1 + 0.10(-2/3) = -1.066667.}$$

Energy quadrature gives $u_- = -1.071797$ and $\Omega = 0.995498$. The approximation captures the mean shift and softening; its turning-point error reflects omitted higher waveform orders.

# The two nonlinear terms enter at different orders

Can a spring with positive cubic stiffness still soften? Use the same release $R$ and the same scaled displacement $u = x/R$ as before:

$$m\ddot{x} + kx + k_2x^2 + k_3x^3 = 0, \qquad \tau = \omega_0 t, \quad \omega_0 = \sqrt{k/m}.$$

Divide by $kR$ and reuse the two dimensionless strengths already introduced:

$$\boxed{u'' + u + \delta u^2 + \varepsilon u^3 = 0}, \qquad \delta = \frac{k_2 R}{k}, \quad \varepsilon = \frac{k_3 R^2}{k}.$$

As $R$ becomes small, $\delta$ is proportional to $R$ while $\varepsilon$ is proportional to $R^2$. The quadratic term first changes the waveform; its feedback and the cubic term change frequency at order $R^2$.

# Find the frequency change from the resonant cosine

From 2.5, $u_1 = \cos\phi$ and $u_2 = -1/2 + (1/3)\cos\phi + (1/6)\cos 2\phi$. The correction at order $R^2$ has three contributions to $\cos\phi$:

$$\cos\phi\cos 2\phi = \tfrac{1}{2}(\cos 3\phi + \cos\phi) \quad \Longrightarrow \quad [u_1 u_2]_{\cos\phi} = -\tfrac{1}{2} + \tfrac{1}{12} = -\tfrac{5}{12}.$$

$$\begin{aligned}
\text{Quadratic feedback:} \quad & -2\delta^2 u_1 u_2 && \longrightarrow \quad +\tfrac{5}{6}\delta^2\cos\phi, \\
\text{Cubic term:} \quad & -\varepsilon u_1^3 && \longrightarrow \quad -\tfrac{3}{4}\varepsilon\cos\phi, \\
\text{Frequency change:} \quad & -2(\Omega-1)u_{1,\phi\phi} && \longrightarrow \quad +2(\Omega-1)\cos\phi.
\end{aligned}$$

A periodic correction needs the sum of these resonant coefficients to vanish:

$$2(\Omega-1) + \frac{5}{6}\delta^2 - \frac{3}{4}\varepsilon = 0 \quad \Longrightarrow \quad \boxed{\Omega \approx 1 + \frac{3}{8}\varepsilon - \frac{5}{12}\delta^2.}$$

The cubic contribution has the sign of $k_3$. The quadratic contribution is negative for either sign of $k_2$.

# Frequency and displacement in physical units

Insert $\delta = k_2 R/k$ and $\varepsilon = k_3 R^2/k$ into $\omega = \omega_0 \Omega$:

$$\boxed{\omega \approx \omega_0 \left[1 + \left(\frac{3k_3}{8k} - \frac{5k_2^2}{12k^2}\right) R^2\right].}$$

For the displacement, retain the first quadratic correction from 2.5. The cubic correction to the waveform is one order smaller in $R$:

$$x \approx R\cos\phi + \frac{k_2 R^2}{k}\left(-\frac{1}{2} + \frac{1}{3}\cos\phi + \frac{1}{6}\cos 2\phi\right), \qquad \phi \approx \omega t.$$

At $\phi = 0$ the correction sums to zero, so $x(0) = R$. Averaging the cosines gives $\bar{x} \approx -k_2 R^2/(2k)$.

The frequency formula retains terms proportional to $R^2$; the displayed waveform retains displacement terms through $R^2$.

# Worked example: positive cubic stiffness, net softening

$$m = 1 \text{ kg}, \quad k = 100 \text{ N/m}, \quad k_2 = 200 \text{ N/m}^2, \quad k_3 = 100 \text{ N/m}^3, \quad R = 0.050 \text{ m}.$$

Here $\omega_0 = \sqrt{100/1} = 10$ rad/s, $\delta = 200(0.050)/100 = 0.10$, and $\varepsilon = 100(0.050)^2/100 = 0.0025$.

$$\frac{3}{8}\varepsilon = 0.00093750, \qquad -\frac{5}{12}\delta^2 = -0.00416667.$$

$$\boxed{\omega \approx 10(1 - 0.00322917) = 9.9677083 \text{ rad/s}.}$$

Quadratic feedback outweighs positive cubic stiffness at this release. The first waveform correction gives $\bar{x} \approx -0.002500$ m and $x_- \approx -R(1 + 2\delta/3) = -0.053333$ m.

# When do the leading stiffness effects cancel?

The coefficient of $R^2$ in the frequency formula determines the initial direction of the backbone:

$$\frac{3k_3}{8k} > \frac{5k_2^2}{12k^2} : \text{ hardening}, \qquad \frac{3k_3}{8k} < \frac{5k_2^2}{12k^2} : \text{ softening}.$$

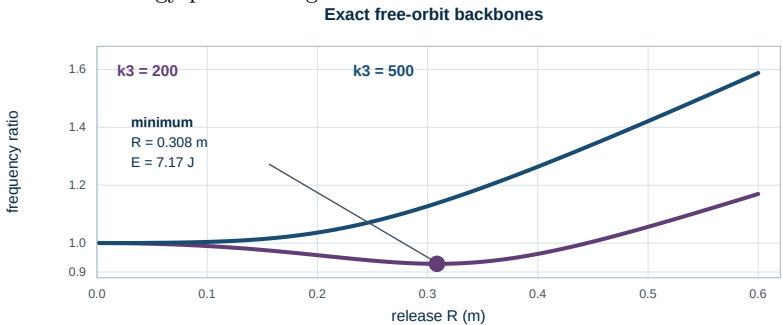
At the boundary, set the two terms equal. Solving for $k_3$ and inserting the example's $k = 100$ and $k_2 = 200$ gives

$$\boxed{k_3 = \frac{10k_2^2}{9k} = \frac{10(200)^2}{9(100)} = 444.444 \text{ N/m}^3.}$$

Cancellation of the $R^2$ term does not prove that every amplitude has the same frequency. Higher-order terms remain.

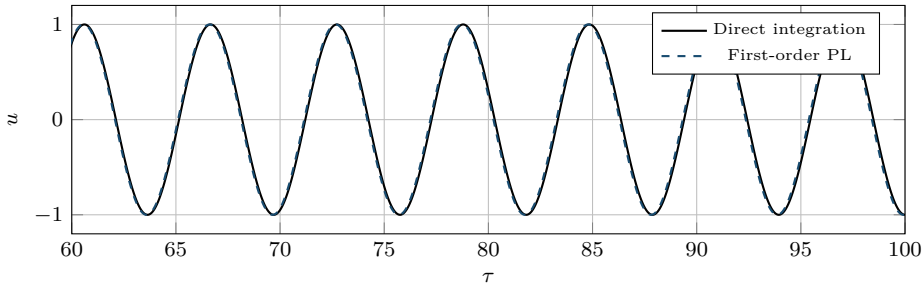
# Can the backbone change direction as energy rises?

Keep $m = 1$ kg, $k = 100$ N/m and $k_2 = 200$ N/m$^2$; compare $k_3 = 200$ and 500 N/m$^3$. For each release, set $E = U(R)$ and use the exact energy-period integral from 2.1.




For $k_3 = 200$, frequency first falls to $\omega/\omega_0 = 0.928$ at $R = 0.308$ m ($E = 7.17$ J), then rises as energy increases. For $k_3 = 500$ it rises over the plotted range. The $R^2$ formula predicts only the initial direction.

# Can a small frequency error matter?




For $\varepsilon = 0.10$, $\Delta\Omega = 0.000783093$. By $\tau = 100$, the phase difference is approximately 0.0783093 rad. PL does not remove errors from every omitted order.

# Validate frequency against an independent reference

For the same cubic model and release $u(0) = 1$, $u'(0) = 0$, evaluate the energy-period integral from 2.2 numerically, then use $\Omega_{\mathrm{quad}} = 2\pi/T_\tau$. Compare it with $\Omega_{\mathrm{PL}} = 1 + 3\varepsilon/8$.

| $\varepsilon$ | Quadrature $\Omega$ | First-order PL | Error (%) |
|---|---|---|---|
| 0.02 | 1.00746750 | 1.00750000 | 0.003226 |
| 0.10 | 1.03671691 | 1.03750000 | 0.075536 |
| 0.30 | 1.10601824 | 1.11250000 | 0.586045 |
| -0.10 | 0.96163744 | 0.96250000 | 0.089697 |
| -0.30 | 0.87878975 | 0.88750000 | 0.991164 |

Error is $100|\Omega_{\mathrm{PL}} - \Omega_{\mathrm{quad}}|/\Omega_{\mathrm{quad}}$. Both calculations use the same cubic model, scaled time and release $R$.

# Check the period, conservation and asymptotic order

## Three independent checks

1. Integrate $u' = v$, $v' = -u - \varepsilon u^3$. Consecutive downward crossings of $u = 0$ are one period apart. Refine the time step.

$$\mathcal{E} = \tfrac{1}{2}v^2 + \tfrac{1}{2}u^2 + \tfrac{1}{4}\varepsilon u^4, \quad \Delta_E = \max_{\tau} \frac{|\mathcal{E}(\tau) - \mathcal{E}(0)|}{|\mathcal{E}(0)|}.$$

2. Compare the measured period with converged energy quadrature; track energy drift throughout the record.

3. Substitute the PL waveform into the equation. A first-order cubic approximation should leave a residual of order $\varepsilon^2$ at fixed phase.

If the residual is $C\varepsilon^2 + O(\varepsilon^3)$, replacing $\varepsilon$ by $\varepsilon/2$ gives approximately one quarter as much residual.

A factor near four. For $0.01 \to 0.005$, the calculated ratio is 4.00214. Residual, frequency error, energy drift and long-time phase error measure different things.

# Solved practice: a softening oscillator

$$m = 2 \text{ kg}, \quad k = 50 \text{ N/m}, \quad k_3 = -200 \text{ N/m}^3, \quad R = 0.10 \text{ m}.$$

Scale the spring model as in 2.4. Its linear frequency and nonlinear strength are

$$\omega_0 = \sqrt{50/2} = 5 \text{ rad/s}, \qquad \varepsilon = \frac{-200(0.10)^2}{50} = -0.0400.$$

The frequency ratio is $\Omega \approx 1 + 3(-0.0400)/8 = 0.9850$, so

$$\boxed{\omega \approx 5(0.9850) = 4.925 \text{ rad/s}.}$$

In the release-preserving waveform, $\varepsilon/32 = -0.00125$ and $\phi = \omega t$:

$$x \approx 0.10[\cos\phi - 0.00125(\cos 3\phi - \cos\phi)] \text{ m}, \quad \phi = 4.925t.$$

Fundamental: 0.100125 m; signed third harmonic: $-0.000125$ m. Their sum is $R$.

# Softening practice: check the energy barrier

$$U(x) = \tfrac{1}{2}kx^2 - \tfrac{1}{4}|k_3|x^4.$$

The nonzero stationary points solve $U'(x) = x(k - |k_3|x^2) = 0$, hence $x_s = \pm\sqrt{k/|k_3|}$. Compare their potential energy with the release energy $E = U(R)$.

$$x_s = \pm\sqrt{k/|k_3|} = \pm 0.50 \text{ m},$$

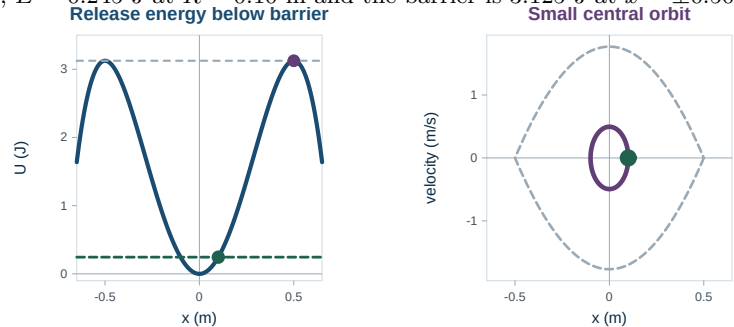
$$U(x_s) = \frac{k^2}{4|k_3|} = \frac{50^2}{4(200)} = 3.125 \text{ J},$$

$$\boxed{E = U(0.10) = \tfrac{1}{2}(50)(0.10)^2 - \tfrac{1}{4}(200)(0.10)^4 = 0.245 \text{ J}.}$$

The release lies inside the central well and below the barrier. The potential is not globally confining, but this small central oscillation is bounded. Near a separatrix, small-amplitude perturbation is not reliable.

# The softening example stays inside its energy barrier

For the preceding example, $E = 0.245$ J at $R = 0.10$ m and the barrier is 3.125 J at $x = \pm 0.50$ m.




Green marks release energy and the purple central orbit. Gray marks the barrier and its phase contour. This recalls Week 1; the Week 2 frequency estimate describes the small bounded orbit.

# Exit check: what should we be able to explain?

### Why no first-order quadratic frequency shift?

The square of the leading cosine contains a mean and a second harmonic, but no fundamental.

### Why does quadratic stiffness change frequency later?

The first-order waveform feeds a fundamental harmonic into the next-order equation.

### What removes the secular term?

Choose the frequency correction so that the resonant Fourier coefficient vanishes.

### What must be checked before applying a formula?

The release $R$, smallness of the nonlinear terms, model truncation and the time interval of interest.

Next week: multiple scales, averaging and conservative applications. Here $\varepsilon = k_3 R^2/k$ measures cubic strength; next week a small parameter counts amplitude, so the same frequency shift appears at second amplitude order.

# Sources for the methods and examples

**[1] A. H. Nayfeh and D. T. Mook**
*Nonlinear Oscillations*. Wiley, 1979.
Secs. 2.3.1–2.3.2: regular expansion, secular terms and Lindstedt–Poincaré method. Sec. 2.4.1: pendulum. Combined stiffness: Eqs. (2.3.30)–(2.3.34).

**[2] O. Thomas**
"Understanding, computing and identifying the nonlinear dynamics of elastic and piezoelectric structures thanks to nonlinear modes." In *Model Order Reduction for Design, Analysis and Control of Nonlinear Vibratory Systems*, 2024.
Fig. 4.4: the pendulum's free-oscillation backbone and its changing waveform; the driven-response curves in that figure serve as a comparison. DOI: 10.1007/978-3-031-67499-0_4.